

Workshop ‘Spatio-temporal modeling for ecology’

Stream network activity

Some (bibliographic) context

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Catchment-scale stream network spatio-temporal models, applied to the freshwater stages of a diadromous fish species, longfin eel (*Anguilla dieffenbachii*)

Anthony R. Charsley^{a,*}, Arnaud Grüss^a, James T. Thorson^b, Merrill B. Rudd^c,
Shannan K. Crow^d, Bruno David^e, Erica K. Williams^a, Simon D. Hoyle^f

Ecological Applications, 28(7), 2018, pp. 1782–1796

Published 2018. This article is a U.S. Government work and is in the public domain in the USA.

A geostatistical state-space model of animal densities for stream networks

DANIEL J. HOCKING,^{1,4} JAMES T. THORSON,² KYLE O'NEIL,³ AND BENJAMIN H. LETCHER³

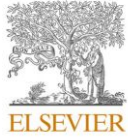
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³Leetown Science Center, S.O. Conte Anadromous Fish Research Laboratory, U.S. Geological Survey, One Migratory Way, Turners Falls Massachusetts 01376 USA

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A geostatistical state-space model of animal densities for stream networks

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Overall framework

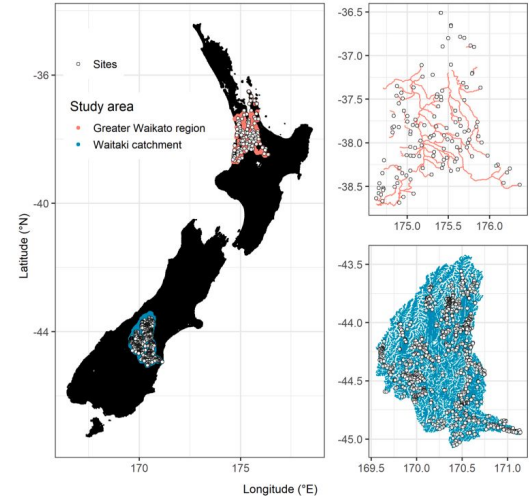
Case study: Longfin eel (*Anguilla dieffenbachii*), New Zealand

Ecological objective: estimating spatio-temporal changes in longfin eel populations in the freshwater environment

Data: Presence absence / Count data of eels, various fishing methods

- + Covariates (e.g. mean flow, distance to coast, mean elevation)

Statistical model: VAST + adaptation to stream network



Modeling framework | VAST

Linear predictor for observation i — $p_1(i) = \underbrace{\mu_{\beta_1}(c_i) + \sum_{f=1}^{n_{\beta_1}} L_{\beta_1}(c_i, f) \beta_1(t_i, f)}_{\text{Temporal variation}} + \underbrace{\sum_{f=1}^{n_{\omega_1}} L_{\omega_1}(c_i, f) \omega_1(s_i, f)}_{\text{Spatial variation}} + \underbrace{\sum_{f=1}^{n_{\epsilon_1}} L_{\epsilon_1}(c_i, f) \epsilon_1(s_i, f, t_i)}_{\text{Spatio-temporal variation}}$

Category (e.g. species) — $\mu_{\beta_1}(c_i)$

Number of temporal effects — n_{β_1}

Number of spatial effects — n_{ω_1}

Number of spatio-temporal effects — n_{ϵ_1}

$+ \underbrace{\sum_{p=1}^{n_p} \gamma_1(c_i, t_i, p) X(s_i, t_i, p)}_{\text{Habitat covariates}} + \underbrace{\sum_{k=1}^{n_k} \lambda_1(k) Q(i, k)}_{\text{catchability covariates}}$

Number of habitat covariates — n_p

Number of catchability covariates — n_k

Modeling framework | VAST

Linear predictor for observation i

$$p_1(i) = \underbrace{\mu_{\beta_1}(c_i) + \sum_{f=1}^{n_{\beta_1}} L_{\beta_1}(c_i, f) \beta_1(t_i, f)}_{\text{Temporal variation}} + \underbrace{\sum_{f=1}^{n_{\omega_1}} L_{\omega_1}(c_i, f) \omega_1(s_i, f)}_{\text{Spatial variation}} + \underbrace{\sum_{f=1}^{n_{\epsilon_1}} L_{\epsilon_1}(c_i, f) \epsilon_1(s_i, f, t_i)}_{\text{Spatio-temporal variation}}$$

Category (e.g. species) $\rightarrow \mu_{\beta_1}(c_i)$

Number of temporal effects $\rightarrow n_{\beta_1}$

Number of spatial effects $\rightarrow n_{\omega_1}$

Number of spatio-temporal effects $\rightarrow n_{\epsilon_1}$

$$+ \underbrace{\sum_{p=1}^{n_p} \gamma_1(c_i, t_i, p) X(s_i, t_i, p)}_{\text{Habitat covariates}} + \underbrace{\sum_{k=1}^{n_k} \lambda_1(k) Q(i, k)}_{\text{catchability covariates}}$$

Number of habitat covariates $\rightarrow n_p$

Number of catchability covariates $\rightarrow n_k$

Temporal, spatial and spatio-temporal **random effects**

The diagram illustrates the VAST modeling framework equation. The main equation is: $p_1(i) = \mu_{\beta_1}(c_i) + \sum_{f=1}^{n_{\beta_1}} L_{\beta_1}(c_i, f) \beta_1(t_i, f) + \sum_{f=1}^{n_{\omega_1}} L_{\omega_1}(c_i, f) \omega_1(s_i, f) + \sum_{f=1}^{n_{\epsilon_1}} L_{\epsilon_1}(c_i, f) \epsilon_1(s_i, f, t_i) + \sum_{p=1}^{n_p} \gamma_1(c_i, t_i, p) X(s_i, t_i, p) + \sum_{k=1}^{n_k} \lambda_1(k) Q(i, k)$. Annotations include: 'Linear predictor for observation i' pointing to $p_1(i)$; 'Category (e.g. species)' pointing to $\mu_{\beta_1}(c_i)$; 'Number of temporal effects' pointing to n_{β_1} ; 'Number of spatial effects' pointing to n_{ω_1} ; 'Number of spatio-temporal effects' pointing to n_{ϵ_1} ; 'Temporal variation' bracket under the first three terms; 'Spatial variation' bracket under the fourth term; 'Spatio-temporal variation' bracket under the fifth term; 'Habitat covariates' bracket under the sixth term; 'catchability covariates' bracket under the seventh term; 'Number of habitat covariates' pointing to n_p ; 'Number of catchability covariates' pointing to n_k ; and 'Temporal, spatial and spatio-temporal random effects' pointing to the random effect terms $\beta_1, \omega_1, \epsilon_1$.

Modeling framework | VAST

Linear predictor for observation i

$$p_1(i) = \underbrace{\mu_{\beta_1}(c_i) + \sum_{f=1}^{n_{\beta_1}} L_{\beta_1}(c_i, f) \beta_1(t_i, f)}_{\text{Temporal variation}} + \underbrace{\sum_{f=1}^{n_{\omega_1}} L_{\omega_1}(c_i, f) \omega_1(s_i, f)}_{\text{Spatial variation}} + \underbrace{\sum_{f=1}^{n_{\epsilon_1}} L_{\epsilon_1}(c_i, f) \epsilon_1(s_i, f, t_i)}_{\text{Spatio-temporal variation}}$$

Category (e.g. species) Number of temporal effects Number of spatial effects Number of spatio-temporal effects

$$+ \underbrace{\sum_{p=1}^{n_p} \gamma_1(c_i, t_i, p) X(s_i, t_i, p)}_{\text{Habitat covariates}} + \underbrace{\sum_{k=1}^{n_k} \lambda_1(k) Q(i, k)}_{\text{catchability covariates}}$$

Number of habitat covariates Number of catchability covariates

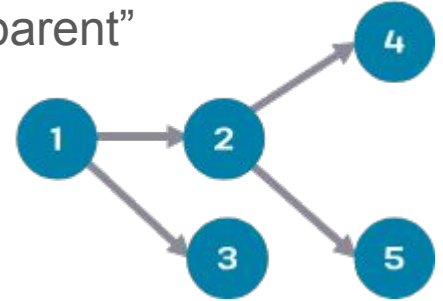
Loading terms
Relate the temporal, spatial and spatio-temporal effects to categories/species

Temporal, spatial and spatio-temporal **random effects**

Modeling framework | Modelling spatio-temporal dependence

Ornstein-Uhlenbeck process

Define the acyclic graph of upstream “child” and downstream “parent”



Modeling framework | Modelling spatio-temporal dependence

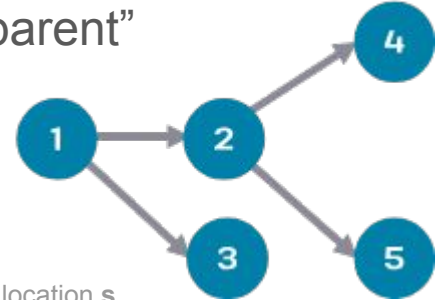
Ornstein-Uhlenbeck process

Define the acyclic graph of upstream “child” and downstream “parent”

$$\omega(s) | \omega(s_{\{parent\}}) \sim \text{Normal}(\rho_s(s) \times \omega(s_{\{parent\}}), \sigma_s^2(s))$$

Expected spatial correlation
between points in the stream network

Variance for spatial correlation for location $\mathbf{s}$,
conditioned upon the value for parent node $\mathbf{s}_{parent}$



Modeling framework | Modelling spatio-temporal dependence

Ornstein-Uhlenbeck process

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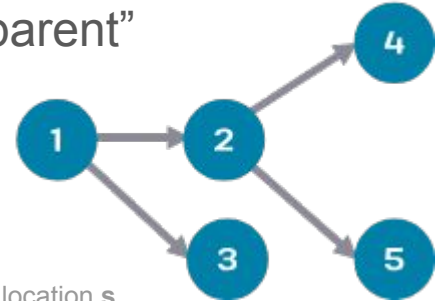
Variance for spatial correlation for location $\mathbf{s}$,
conditioned upon the value for parent node $\mathbf{s}_{parent}$

$$\sigma_s^2(s) = \frac{\sigma_\tau^2}{2\theta_\tau} \left(1 - e^{-2\theta_\tau |s - s_{parent}|} \right)$$

Exponential **rate of decorrelations**
between child and parent nodes,
⇒ Larger values represent faster decorrelation

Asymptotic variance for
two infinitely distant nodes

Distance between location $\mathbf{s}$
and parent node $\mathbf{s}_{parent}$



Modeling framework | Modelling spatio-temporal dependence

Ornstein-Uhlenbeck process

Define the acyclic graph of upstream “child” and downstream “parent”

$$\omega(s) | \omega(s_{\{parent\}}) \sim \text{Normal}(\rho_s(s) \times \omega(s_{\{parent\}}), \sigma_s^2(s))$$

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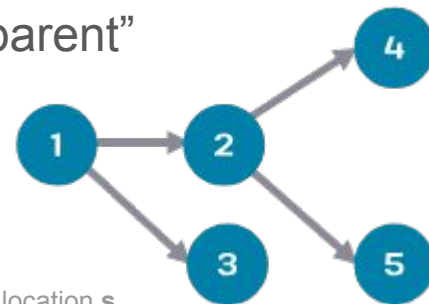
$$\rho_s(s) = e^{-\theta_\tau |s - s_{parent}|}$$

$$\sigma_s^2(s) = \frac{\sigma_\tau^2}{2\theta_\tau} \left(1 - e^{-2\theta_\tau |s - s_{parent}|} \right)$$

Asymptotic variance for
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Distance between location $\mathbf{s}$
and parent node $\mathbf{s}_{parent}$

Exponential rate of decorrelations
between child and parent nodes,
⇒ Larger values represent faster decorrelation



Modeling framework | Modelling spatio-temporal dependence

Ornstein-Uhlenbeck process

Define the acyclic graph of upstream “child” and downstream “parent”

Then define the **precision matrix**:

$$Q_{stream}(s, s_{parent}) = Q_{stream}(s_{parent}, s) = \frac{-e^{-\theta_{\tau}|s-s_{parent}|}}{1-e^{-2\theta_{\tau}|s-s_{parent}|}}$$

Precision $\Rightarrow$ represents the **conditional dependence** of child nodes given parent nodes

Exponential rate
of decorrelations

Distance between location s
and parent node s_{parent}

Modeling framework | Modelling spatio-temporal dependence

Ornstein-Uhlenbeck process

Define the acyclic graph of upstream “child” and downstream “parent”

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Precision $\Rightarrow$ represents the **conditional dependence** of child nodes given parent nodes

Exponential rate of decorrelations

Distance between location s and parent node s_{parent}

Diagonal terms of the precision matrix

$$Q_{stream}(s, s) = 1 + \sum_{s' \in S} \frac{e^{-2\theta_{\tau}|s-s'|}}{1-e^{-2\theta_{\tau}|s-s'|}}$$

Set of child and parent nodes that are **adjacent** to node s

Modeling framework | Modelling spatio-temporal dependence

Ornstein-Uhlenbeck process

Define the acyclic graph of upstream “child” nodes and downstream “parent”

Then define the **precision matrix**

$$Q_{stream}(s, s_{parent}) = \frac{-e^{-\theta_{\tau}|s-s_{parent}|}}{1-e^{-2\theta_{\tau}|s-s_{parent}|}}$$

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Set of child and parent nodes that are **adjacent** to node s

Markovian properties

Modeling framework | Modelling spatio-temporal dependence

Ornstein-Uhlenbeck process

Define the acyclic graph of upstream “child” and downstream “parent”

Then define the **precision matrix**.

Extension to time through a first-order autocorrelation process (not described here).

Modeling framework | Modelling spatio-temporal dependence

Ornstein-Uhlenbeck process

Define the acyclic graph of upstream “child” and downstream “parent”

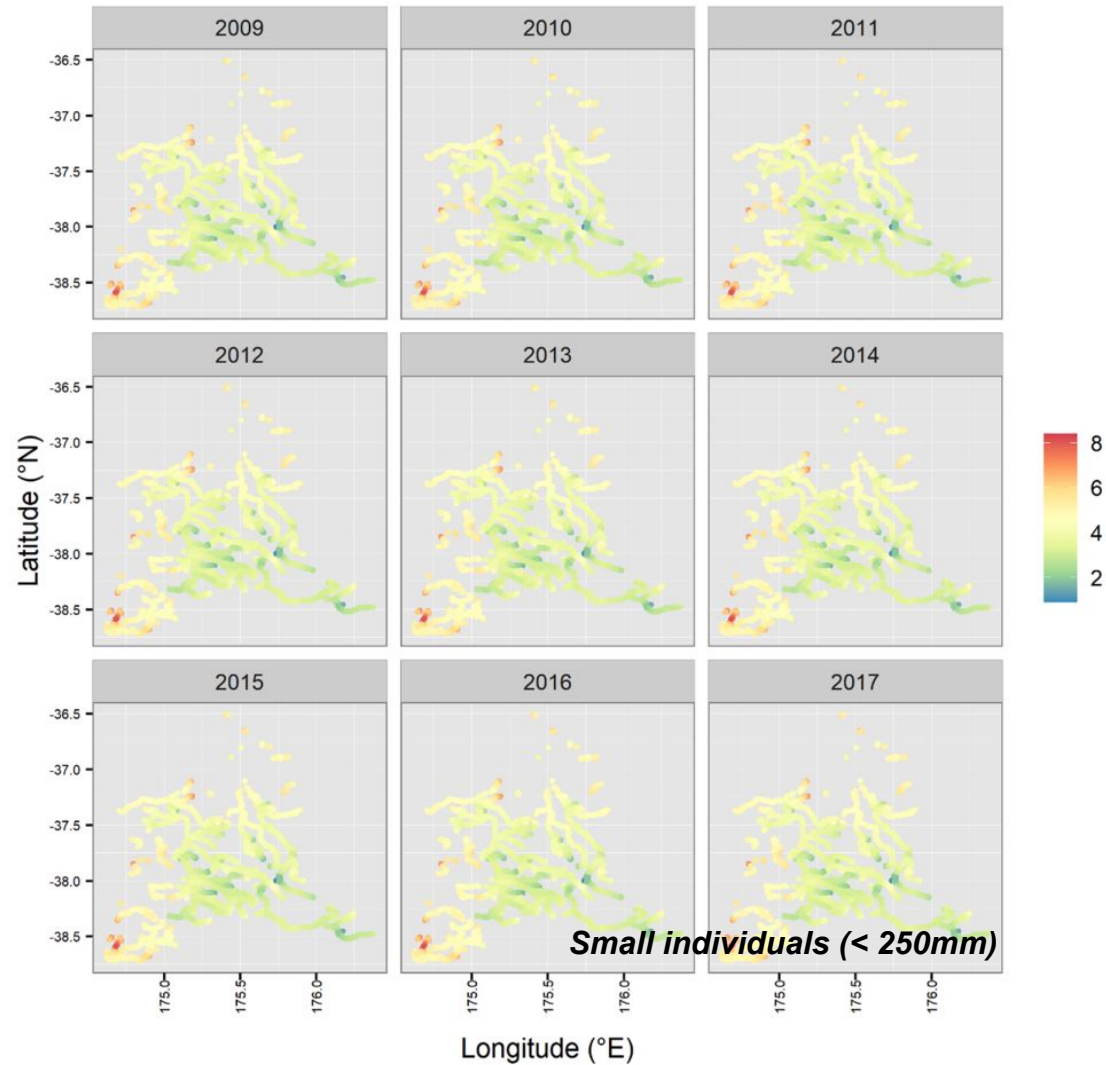
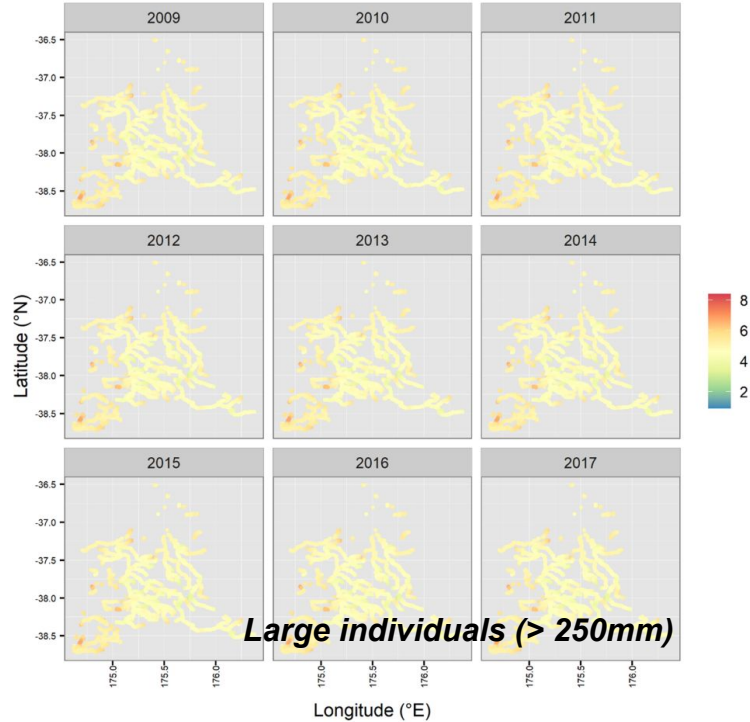
Then define the **precision matrix**.

Extension to time through a first-order autocorrelation process (not described here).

Inference realized with TMB (standard for VAST/tinyVAST).

Some results

Map of predicted count/km



Our case study

Case study: Freshwater fish in the Loire River basin (France)

Ecological objective: Estimating spatio-temporal changes in fish populations in the freshwater environment

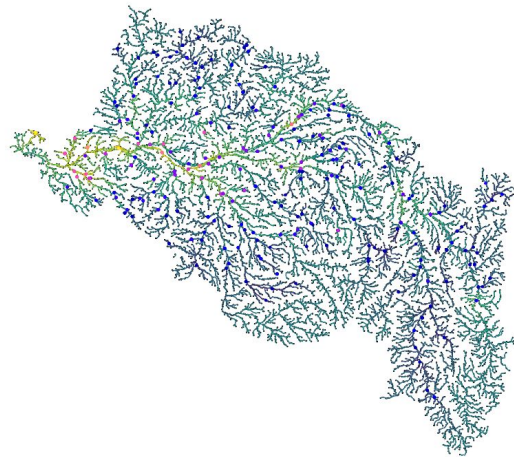
Data: Count data of freshwater fish in the Loire River bassin, 1990-2024 (ASPE database, OFB)

+ Covariates (e.g. distance to coast, elevation)

Statistical model: tinyVAST with adaptation to stream network



European eel spatial distribution in the Loire Bassin



Codes and questions to address

Lien [codes](#)

Lien [données](#)

Lien [tutoriels tinyVAST](#)

Lien [présentation](#)

Checking the network

Adding environmental covariates → need an interpolation step for prediction

Adding additional species (e.g. ~~TEF~~ truite ; SPI – spiralin ; CHE – chevesne ;

Other questions ?